$A \sim$ poisson, λ_A per dygn $\rightarrow X_A$ = # fel et år

$B \sim$ poisson, λ_B per dygn $\rightarrow X_B$ = # fel et år

a)

$\lambda_A = \frac{3}{73} = 0{,}0411$ $\wedge$ $\lambda_B = \frac{1}{73} = 0{,}0137$

$E[X_A] = ?$

$\lambda_A = E[\#\text{ fel en dag}] \Rightarrow X_A : \mu_A = 365 \cdot \lambda_A$

$365 \cdot \frac{3}{73} = 73 \cdot 5 \cdot \frac{3}{73} = \underline{\underline{15}}$

$P(X_A \leq 15) = ?$

$\mu_A = 15$, $x = 15$ ser i blå bok s. 20:

$P(X_A \leq 15) = \underline{\underline{0{,}5681}}$

$P(X_A \leq 15 \ , X_B \leq 10) = ?$ Antar $= P(X_A \leq 15 \cap X_B \leq 10)$

$E[X_B] = 365 \cdot \frac{1}{73} = 73.5 \cdot \frac{1}{73} = \underline{5}$

Teori:

$P(A \cap B) \overset{\text{uavh.}}{\Longrightarrow} P(A) \times P(B)$

$P(X_B \leq 10), \ \mu_B = 5, \ x = 10 \Rightarrow 5.18 = \underline{0,9863}$

$P(X_A \leq 15 \cap X_B \leq 10) = 0,5681 \times 0,9863$

$= 0,55009 \approx \underline{0,55}$

$P(X_A \leq 15 \mid X_B \leq 10) \overset{\text{uavh.}}{\Longrightarrow} P(X_A \leq 15) = \underline{0,5681}$

Teori

$P(A|B) = \frac{P(A \cap B)}{P(B)} \overset{\text{uavh.}}{\Longrightarrow} \frac{P(A)P(B)}{P(B)}$

b) $Z = X_A + X_B$

$f_Z(z) =$ poisson?

Teori $M_X(t) = e^{\mu_i(e^t - 1)}$

$$M_Z(t) = E[e^{tZ}] = E[e^{t(X_A + X_B)}] = E[e^{tX_A} e^{tX_B}]$$

$$= E[e^{tX_A}] \cdot E[e^{tX_B}]$$

$$= e^{\mu_A(e^t-1)} \cdot e^{\mu_B(e^t-1)} = e^{(\mu_A+\mu_B)(e^t-1)}, \text{ si } \mu_A + \mu_B = \mu_Z$$

$\Rightarrow M_Z(t) = e^{\mu_Z(e^t-1)}$ | poissonfördelning ☑

Q.E.D

c) $f_{X_A|Z}(x_A|z) = P(X_A = x_A | Z = z)$

$$= \frac{P(X_A = x_A \cap Z = z)}{P(Z = z)}$$

$Z = X_A + X_B$

$X_B = Z - X_A \Rightarrow x_B = z - x_A$

$$\frac{P(X_A = x_A \wedge X_B = z - x_A)}{P(Z = z)}$$

← beholder $Z = z$ for å ikke fjerne z

Teori

f poisson: $f_x = \frac{\nu^x}{x!} e^{-\nu}$, $x = 365$ her

$$\frac{\left(\frac{\mu_A^{x_A}}{x_A!} e^{-\mu_A}\right) \cdot \frac{(\mu_B)^{z-x_A}}{(z-x_A)!} \cdot e^{-\mu_B}}{\frac{(\mu_A+\mu_B)^z}{z!} e^{-(\mu_A+\mu_B)}} = \frac{\mu_A^{x_A} \mu_B^{z-x_A}}{x_A!\,(z-x_A)!} \cdot \frac{z!}{(\mu_A+\mu_B)^z} e^{-\mu_A-\mu_B+\mu_A+\mu_B}$$

teller (under $\frac{\mu_A^{x_A}\mu_B^{z-x_A}}{x_A!(z-x_A)!}$), nevner (under $\frac{z!}{(\mu_A+\mu_B)^z}$)

$$= \frac{z!}{x_A!(z-x_A)!} \frac{\mu_A^{x_A} \cdot \mu_B^{z} \cdot \mu_B^{-x_A}}{(\mu_A+\mu_B)^z}$$

$$\alpha^z = \alpha^{x_A} \cdot \alpha^{z-x_A}$$

$$= \binom{z}{x_A} \left(\frac{\mu_A}{\mu_A+\mu_B}\right)^{x_A} \left(\frac{\mu_B}{\mu_A+\mu_B}\right)^{z-x_A}$$

$\binom{n}{x} p^x (1-p)^{n-x}$

$n = z$, $x = x_A$, $p = \frac{\mu_A}{\mu_A+\mu_B}$

ser fra denne, en binomisk fordeling!

Oppg 2 X = fødselsvekt i gram

$X \sim N(3500, 570^2)$

a) $P(X \geq 3000) = 1 - P(X \leq 3000)$

Sentralgrensetreorem

Si $Z = \dfrac{\bar{X} - \mu}{\sqrt{\frac{\sigma^2}{n}}}$ der $\bar{X}$ = gjennomsnittsvekt og n = # nyfødte

$\lim_{n \to \infty} Z \approx N(0,1)$, $\bar{X} = \sum_{i=1}^{n} x_i \Rightarrow \bar{X} = n \cdot X, \sim N(n\mu, n\sigma^2)$

$1 - P(X \leq 3000) = 1 - P\left(Z \leq \dfrac{3000 - \mu}{\sqrt{\sigma^2/n}}\right)$

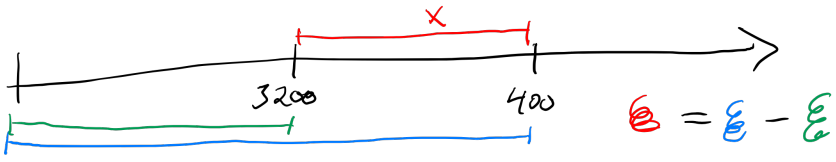
$\mu = 3500$, $\sigma = 570$, $n = 1$

$\Rightarrow 1 - P(Z \leq -0{,}8772) = 1 - \Phi(-0{,}8772)$

ibla bok!

$= 1 - 0{,}1922 = 0{,}8078$

$$P(3200 \leq X < 4000) = ?$$



$$= P(X < 4000) - P(X \leq 3200)$$

(continuidade)

$$= P\left(Z \leq \frac{4000 - 3500}{570}\right) - P\left(-\frac{300}{570}\right)$$

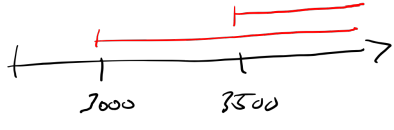
$$= \Phi(0{,}8772) - \Phi(-0{,}5263)$$

$$= 0{,}8078 - 0{,}3015$$

$$= 0{,}5063$$

$$P(X \geq 3500 \mid X \geq 3000) = ?$$

$$= \frac{P(X \geq 3500 \cap \bar{X} \geq 3000)}{P(\bar{X} > 3000)}$$



$$= \frac{P(X \geq 3500)}{P(X \geq 3000)} = \frac{0{,}50}{0{,}8078} \leftarrow N(3500, \sigma^2)$$

$$= 0{,}6190$$

$$P(X \leq 4000 \mid X \geq 3200) = ?$$

$$= \frac{P(X \leq 4000 \cap X \geq 3200)}{P(X \geq 3200)} = \frac{P(3200 \leq X \leq 4000)}{P(X \geq 3200)}$$

$$= \frac{0{,}5063}{1 - \Phi\left(-\frac{300}{570}\right)} = \frac{0{,}5063}{0{,}6985}$$

$$= 0{,}7248$$

b) $P(\bar{X} < c) = 0{,}01$

$\bar{Y}_{n=100}$ der $P(\bar{X} < c)$

$c = ?$ Revers sentralgrenseteorem!

$\Phi(\alpha) = 0{,}01$, vi vet at $c < \mu$

$\Rightarrow \alpha < 0$

Ser i tabell

at $\Phi(-2{,}32) = 0{,}0102 \Rightarrow \alpha \approx -2{,}32$

$\alpha = -2{,}32 = \dfrac{c - 3500}{570}$ | $n=1$, regner ikke på $\bar{Y}$

$c = -570 \cdot 2{,}32 + 3500$

$c = 3500 - 1322$

$\underline{c \approx 2178\ g}$

$Y \sim \text{binom}(100, 0.01)$?

binom: $\binom{n}{x} p^x (1-p)^{n-x}$

Antagelser: Bernoulli's forsøksrekke

- rekkefølgen på de nyfødte må være uavhengige (rimelig)
- 2 mulige utfall: undervektig eller ikke
- Lik sannsynlighet for undervektighet hos alle nyfødte (ikke rimelig i virkeligheten)

$P(Y \geq 1)$, siden $np \leq 5$ bruker jeg ikke Normalford.-tilnæring

$$= 1 - P(Y \leq 1) = 1 - F(1) = 1 - \sum_{t=0}^{Y=1} \binom{n}{t} p^t (1-p)^{n-t}$$

$$= 1 - \left[\binom{100}{0} (1-p)^{100} + \underbrace{\binom{100}{1} 0{,}01}_{100 \times 0{,}01} \cdot (0{,}99)^{99} \right]$$

$$= 1 - (0{,}99^{100} + 0{,}99^{99})$$

$$= 1 - 0{,}7358$$

$$= \underline{0{,}2642}$$

$$P(Y \geq 2 \mid Y \geq 1) = \frac{P(Y \geq 2)}{P(Y \geq 1)}$$

$$P(Y \geq 2) = 1 - (0{,}7358 + \binom{100}{2} 0{,}01^2 \cdot 0{,}99^{98})$$

$$= 0{,}2642 - 0{,}1849 = \underline{0{,}07934}$$

$$\frac{0{,}07934}{0{,}2642} = \underline{0{,}3003}$$

Opg 3 / $Z_i \sim \text{poisson}(\lambda)$, hendelse nr i

$X_1 = Z_1$

$X_k = Z_k - Z_{k-1}$, $k \geq 2$, $k \in \mathbb{Z}$

X_i = varhenge, eksponentialfordelte (λ) , $\lambda = \frac{1}{\beta}$

$$P(X_1 \geq 2) = 1 - P(X_1 \leq 2)$$

$$= 1 - F_Z(2) = 1 - e^{-\lambda 2}$$

$$= e^{\lambda \cdot 2} = e^{-1} = \underline{\underline{0,3679}}$$

$$P(X_1 + X_2 \geq 4) = 1 - P(X_2 \leq 4 - X_1)$$

$$= 1 - \sum_{i=1}^{k} P(X_2 \leq 4 - X_1 \mid X_1 = x_i) \cdot P(X_1 = x_i)$$

Kontinuerlig

$$= 1 - \int_0^4 P(X_2 \leq 4 - X_1 \mid X_1 = x_1) \cdot P(X_1 = x_1) \, dx_1$$

$$P(X_1 = x_1) = f_{X_1}(x_1) = \lambda e^{-\lambda x_1}$$

$P(X_2 \leq 4 - x_1 \mid X_1 = x_1)$, har ej $4 - x_1 \in \mathbb{N} \Rightarrow P(X_2 \leq w)$, $w = 4 - x_1$

$$= 1 - e^{-\lambda(4 - x_1)}$$

$$\Rightarrow P(X_1 + X_2 < 4) = 1 - \int_0^4 \left(1 - e^{-\lambda(4 - x_1)}\right) \lambda e^{-\lambda x_1} \, dx_1$$

$$= 1 - \lambda \int_0^4 e^{-\lambda x_1} - e^{-\lambda 4 + \lambda x_1 - \lambda x_1} \, dx_1$$

$$= 1 - \lambda \int_0^4 e^{-\lambda x_1} - e^{-4\lambda} \, dx_1 = 1 - \lambda \left[-\frac{1}{\lambda} e^{-\lambda x_1} \right]_0^4 + 4\lambda e^{-4\lambda}$$

$$= 1 - \lambda \left(-\frac{1}{\lambda} e^{-4\lambda} + \frac{1}{\lambda} \right) + 4\lambda e^{-4\lambda}$$

$$= 1 + e^{-4\lambda} - 1 + 4\lambda e^{-4\lambda}$$

$$= e^{-4\lambda}(1 + 4\lambda)$$

$\lambda = 0{,}5 \Rightarrow \underline{0{,}4060}$

$$P(X_1 + X_2 \geq 4 \mid X_1 \geq 2) = \frac{P(X_1 + X_2 \geq 4 \cap X_1 \geq 2)}{P(X_1 \geq 2)}$$

$$P(X_1 + X_2 \geq 4 \cap X_1 \geq 2) = 1 - \int_2^4 \left(1 - e^{-\lambda(4 - x_1)}\right) \lambda e^{-\lambda x_1} \, dx_1$$

$$= 1 - \lambda \int_2^4 e^{-\lambda x_1} - e^{-4\lambda} \, dx_1 = 1 - \lambda \int_2^4 e^{-\lambda x_1} \, dx_1 + \lambda e^{-4\lambda} \int_2^4 1 \, dx_1$$

$$= 1 - \lambda \int_2^4 e^{-\lambda x_1} \, dx_1 + 2\lambda e^{-4\lambda}$$

$$= 1 - \lambda \left[-\frac{1}{\lambda} e^{-\lambda x_1} \right]_{x_1 = 2}^{4} + 2\lambda e^{-4\lambda}$$

$$= 1 - \lambda\left(-\frac{1}{\lambda}e^{-4\lambda} + \frac{1}{\lambda}e^{-2\lambda}\right) + 2\lambda e^{-4\lambda}$$

$$= 1 + e^{-4\lambda} - e^{-2\lambda} + 2\lambda e^{-4\lambda}$$

$\lambda = 0,5$

$= \underline{0,9028}$

$$P(X_1 + X_2 \geq 4 \mid X_i \geq 2) = \frac{P(X_1 + X_2 \geq 4 \wedge X_1 \geq 2)}{P(X_1 \geq 2) = 0,3679}$$

$1 - P(X_2 \leq 4 - X_1)$

$$= 1 - \sum_{i=2}^{k} P(X_2 \leq 4 - X_1 \mid X_1 = x_i) \cdot P(X_1 = x_i)$$

$P(X_1 + X_2 \leq 4 \wedge X_1 \geq 2)$

$$1 - \frac{0,0972}{0,3679} = \underline{0,7358}$$

Opg 4

$X \sim N(\mu, \sigma^2)$

$Y = aX + b$

$$E[Y] = E[aX + b] = aE[X] + b = a\mu + b$$

$$Var[Y] = Var[aX + b] = a^2 Var[X] + 0 = a^2\sigma^2$$

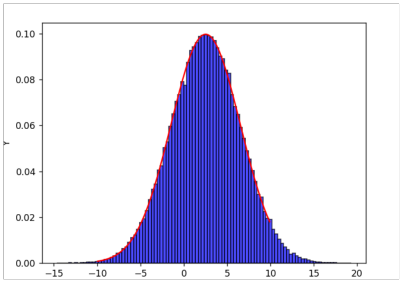
```python
"""

import numpy as np
import matplotlib.pyplot as plt
a = 2
b = 0.5
my = 1
def generateY(n):
    X = np.random.normal(loc = my, scale = 2, size = n)
    Y = a*X + b
    return Y

def f(forvent, s, x):
    nevner = np.sqrt(np.pi*2)*s
    exponent = -((x-forvent)**2)/(2*(s**2))

    return (1/nevner)*np.exp(exponent)

sigma_ny = np.sqrt(a**2*4)
my_ny = a*my + b
w = np.arange(-10,10, 0.1)
plt.plot(w, f(my_ny, sigma_ny, w), 'r-')
plt.hist(generateY(100000), bins=100, density=True, alpha = 0.7, color='b', edgecolor='bla
plt.xlabel('')
plt.ylabel('Y')
plt.show()
```



b) $Y = cX + b \Rightarrow X = \frac{Y-b}{a} = w(y)$

$f_Y(y) = f_X(w(y)) \cdot |w'(y)|$, $w'(y) = \frac{1}{a}$

$$= \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right)} \cdot \left|\frac{1}{a}\right|$$

Oppg 5| 6 pers/stol
3000 pers/time } 500 stoler/time

⟷ = 4 min , 10 min igjen

P(minst én ledig) = 1 - P(ingen) ledig

= 1 - 0,2 = 0,8 per stol

$500 \frac{\text{stoler}}{\text{time}} \cdot (10-4)\text{min} \cdot \frac{1}{60} \text{ min/time}$

hvis stol disse vil hun komme 12:02
= etter ønsket tidspunkt.

50 stoler / 6 min

P(minst én ledig plass på stol på 6 min)

Binomisk fordeling: ledig eller ikke, $p = 0{,}8$

$n = 50$

$$P(Y \geq 1) = 1 - P(Y < 1)$$

$$= 1 - P(Y = 0)$$

$$= 1 - \binom{50}{0} p^0 (1-p)^{n-x}$$

$$= 1 - 0{,}2^{50}$$ $= 0{,}2^{50}$ = ingen ledige på 50 stoler

$$= \underline{\underline{1{,}00}}$$

$\approx 100\%$ ser at dette er lik geometrisk fordeling

Oppg 6) $p_1, p_2 \in \{0, 1\}$

$P(X=0) = (1-p_1)(1-p_2)$

$P(X=1) = p_1 + p_2 - 2p_1p_2$

$P(X=2) = p_1p_2$

} Trinnvis fordelt

a)

$$M_x(t) = E\left[e^{tx}\right] = \sum_x e^{tx} f(x)$$

$$(1-p_1)(1-p_2) + e^t(p_1+p_2-2p_1p_2) + e^{2t} \cdot p_1p_2$$

$$1 - 2p_1p_2 + p_1p_2 + p_1e^t + p_2e^t - 2p_1p_2e^t + p_1p_2e^{2t}$$

$$= \underline{\underline{1 - 2p_1p_2(1+e^t) + p_1p_2(e^{2t}+1)}}$$

$$S_n = \sum_{i=1}^{n} X_i$$

b) $$M_{S_n}(t) = E\left[\sum_{i=1}^{n} e^{tX_i}\right] = E\left[e^{t(X_1 + X_2 + \dots)}\right]$$

$$= E\left[e^{tX_1} e^{tX_2} e^{tX_3} \dots\right] = \prod_{i=1}^{n} E\left[e^{tX_i}\right] \dots$$

$$\left(1 - 2p_1p_2\left(1 + e^{t}\right) + p_1p_2\left(e^{2t} + 1\right)\right)^{n}$$

c) $Y \sim bSnom(n, p_1)$

$$M_Y(t) = E\left[e^{tY}\right] = \sum_{x} \binom{n}{x} p^{x} (1-p)^{n-x}$$